



DSA – Sec A

Lab – 06

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1. **Decimal To Binary** Conversion Using Recursion.

Sample input and output:

Input : 7

Output: 111

Input : 10

Output: 1010

2. Given a string, write a recursive function that checks if the given string is a **palindrome**, else, not a palindrome.

Sample input and output:

Input : malayalam

Output: Yes

Reverse of malayalam is also
malayalam.

Input : max

Output: No

Reverse of max is not max.

3. Sum triangle from array using recursion.

Given an array of integers, print a sum triangle from it such that the first level has all array elements. From then, at each level number of elements is one less than the previous level and elements at the level is be the Sum of consecutive two elements in the previous level.

Sample input and output:

Input : A = {1, 2, 3, 4, 5}

Output: [48]

[20, 28]

[8, 12, 16]

[3, 5, 7, 9]

[1, 2, 3, 4, 5]

Explanation :

Here, [48]
 [20, 28] --> (20 + 28 = 48)
 [8, 12, 16] --> (8 + 12 = 20, 12 + 16 = 28)
 [3, 5, 7, 9] --> (3 + 5 = 8, 5 + 7 = 12, 7 + 9 =
16)
 [1, 2, 3, 4, 5] --> (1 + 2 = 3, 2 + 3 = 5, 3 +
4 = 7, 4 + 5 = 9)

4. Print all **binary Strings** of fixed length k using recursive function

Sample input and output:

Input : 2

Output:

00
01
10
11

Input : 4

Output:

0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111

5. PowerSet Computation

Compute the power set for the given array using recursion.

Sample input and output:

Input : A = {1, 2, 3, 4}

Output: { } { 1 } { 2 } { 1 2 } { 3 } { 1 3 } { 2 3 }
{ 1 2 3 } { 4 } { 1 4 } { 2 4 } { 1 2 4 } { 3 4 } { 1
3 4 } { 2 3 4 } { 1 2 3 4 }